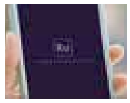
Facial recognition at airports
The TSA has announced plans to use facial recognition technology for Domestic Flights. **https://dgit.in/nov18-76-1**



Adobe's new app for YouTubers
Adobe just launched the Adobe Premier Rush CC, a video editing app designed specifically for online video creators. **https://dgit.in/nov18-76-2**

SAMPLE RETURN MISSIONS TO ASTEROIDS

Humans have only managed one successful mission to an asteroid, and that was a flyby. Now, two ambitious missions in progress plan to grab pieces of these space rocks and bring them back to Earth for further study.

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There has only been one successful mission to an asteroid so far. The planners of the New Horizons mission, did not actively consider any targets on the asteroid belt, to conserve fuel. However, there was an asteroid close enough to observe on the way, called Asteroid 132524 APL. The spacecraft studied the properties of this asteroid, and used it as an opportunity to test out the imaging systems on board. Now, two space agencies are leapfrogging to the next step - missions that will go to an asteroid, gather some of the material, and bring it back to Earth for further study. In June 2018, the Hayabusa 2 mission by JAXA, rendezvoused with 162173 Ryugu, a near Earth carbonaceous asteroid. Studying this type of asteroid will allow scientists to find out more about the formation of the Earth, and maybe even about the origins of life.

HAYABUSA 2

The Hayabusa 2 is a solar powered spacecraft with ion propulsion. Hayabusa 2 is expected to conduct experiments on Ryugu till the end of 2019, after which it will begin its return journey, with a plan to reach home by the end of 2020. There are plans to obtain three samples from various locations on the asteroid. Two of the samples will be obtained through a sampling horn. This horn fires a tantalum bullet to the surface of the asteroid, and captures the ejected material directly. These two are the surface samples. The third sample, will be a subsurface sample. Gathering the third sample involves reaching deeper into the asteroid, to gather material that has not been affected by space weather or heat. To do this, Hayabusa will essentially detonate an explosive on the surface of Ryugu. The explosion is expected to create an artificial crater that is two meters wide. Then Hayabusa 2 will wait for two weeks for all the dust from the explosion to settle down. After that, the spacecraft will visit the new crater to collect its third and final sample.

Hayabusa also has four rovers on board. These rovers will explore the asteroid, and provide additional contextual information for the returned samples. There are three hexagonal Micro Nano Experimental Robot Vehicle for Asteroid (Minerva) rovers on board, which are equipped with cameras, are autonomous, and will hop across the surface of Ryugu like robotic grasshoppers. The rovers have adopted the hopping strategy, considering the low gravity environment on the asteroid. In September, the two Minerva rovers successfully landed on the asteroid. They are equipped with colour cameras, and are in the process of surveying the asteroid, as well as capturing stereoscopic images. The fourth rover, called Mascot was developed by the German Aerospace Center. Mascot had a rather short mission duration of under 17 hours, but managed to pack in a lot of science in that time. After landing on the asteroid on 2 October, it visited the south pole of the asteroid, found some rather large boulders, and found very little regolith, which was a surprise.

OSIRIS-REX

Origins, Spectral Interpretation, Resource Identification, Security-Regolith Explorer (OSIRIS-REx) launched on 8 September 2016, and is on its way to 101955 Bennu. The spacecraft is expected to rendezvous with the asteroid towards the end of the year. On approach, the spacecraft will start mapping the asteroid to choose a suitable site for gathering the sample. This is expected to be an extensive remote sensing campaign. The strategy used to gather the sample is very different from the one used by Hayabusa 2. OSIRIS-Rex will not really land on the asteroid at all. Using the low gravity, it will reduce the altitude of its orbit till it is almost grazing the surface. Over the course of the